Data Structure Assignment 5

Dr. Ranjit Rajak

Array Based Questions

1. Show through appropriate data structure representation of the following 4x4 Sparse Matrix.

$$\begin{pmatrix} 0 & 0 & 11 & 0 \\ 12 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & -25 \end{pmatrix}_{4*4}$$

2. Calculate the address of X[4,3] in a two dimensional array X[1-5,1-4] stored in row major order. Assume the base address to be 1000 and that each element requires 4 words of storage.

3. Draw the vector representation of the sparse matrix of the following matrix:

$$\begin{pmatrix} 0 & 12 & 0 & 0 \\ 0 & 11 & 0 & 0 \\ 9 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}_{4*4}$$

4. Suppose multidimensional arrays A and B are declared using

   A(-2:2, 2:22) and B(1:8, -5:5, -10:5)

   (a) Find the length of each dimension and the number of elements in A and B.

   (b) Consider the element B[3, 3, 3] in B. Find the effective indices E_1, E_2, E_3 and the address of the element, assuming Base(B) = 400 and there are w = 4 words per memory location.

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5. Consider the alphabetized linear array NAME in Fig. 4.23.

	Name
1.	Allen
2.	Clark
3.	Dickens
4.	Edwards
5.	Goodman
6.	Hobbs
7.	Irwin
8.	Klein
9.	Lewis
10.	Morgan
11.	Richards
12.	Scott
13.	Tucker
14.	Walton

(a) Find the number of elements that must be moved if Brown, Johnson and Peters are inserted into NAME at three different times.

(b) How many elements are moved if the three names are inserted at the same time?

(c) How does the telephone company handle insertions in a telephone directory?

6. Consider the linear arrays .
AAA(5:50), BBB(-5:10) and CCC(18).
(a) Find the number of elements in each array.
(b) Suppose Base(AAA) = 300 and w = 4 words per memory cell for AAA. Find the address of AAA[15], AAA[35] and AAA[55].

7. Suppose a company keeps a linear array YEAR(1920:1970) such that YEAR[K] contains the number of employees born in year K. Write a module for each of the following tasks:
(a) To print each of the years in which no employee was born.
(b) To find the number NNN of years in which no employee was born.

(c) To find the number N50 of employees who will be at least 50 years old at the end of the year. (Assume 1984 is the current year.)

(d)To find the number NL of employees who will be at least L years old at the end of the year. (Assume 1984 is the current year.)

8. Suppose a 10-element array A contains the values $a_1, a_2, \ldots, a_{10}$- Find the values in A after each loop.

(a) Repeat for K = 1 to 9:

Set A[K + 1] := A[K].

[End of loop.]

(b) Repeat for K = 9 to 1 by -1:

Set A[K + 1] := A[9].

[End of loop.]

Note that the index K runs from 1 to 9 in part (a) but in reverse order from 9 back to 1 in part (b)

9. Let A be an n x n square matrix array. Write a module which

(a) Finds the number NUM of nonzero elements in A

(b) Finds the SUM of the elements above the diagonal, i.e., elements A[I, J] where I < J

(c) Finds the product PROD of the diagonal elements ($a_{11}, a_{22}, \ldots, a_{nn}$)

10. Consider the linear arrays XXX(-10:10), YYY(1935: 1985), ZZZ(35). (a) Find the number of elements in each array. (b) Suppose Base(YYY) = 400 and w = 4 words per memory cell for YYY. Find the address of YYY[1942], YYY[1977] and YYY[1988].
11. Consider the following multidimensional arrays:

X(-5:5, 3:33) Y(3:10, 1:15, 10:20)

(a) Find the length of each dimension and the number of elements in X and Y.

(b) Suppose Base(Y) = 400 and there are w = 4 words per memory location. Find the effective indices E1, E2, E3 and the address of Y[5, 10, 15] assuming (i) Y is stored in row-major order and (ii) Y is stored in column-major order.

12. An array A contains 25 positive integers. Write a module which

(a) Finds all pairs of elements whose sum is 25.

(b) Finds the number EVNUM of elements of A which are even, and the number ODNUM of elements of A which are odd.

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13. Suppose A is a linear array with n numeric values. Write a procedure

MEAN(A, N, AVE)

which finds the average AVE of the values in A. The arithmetic mean or average x of the values $x_1, x_2, ..., X_n$ is defined by

$$\bar{X} = \frac{X + X_2 + \ldots + X_n}{n}$$

Question Based On Stack

1. Explain underflow and overflow condition of the STACK with proper example.

2. Convert infix to postfix without using STACK.

   i. (A-B) * (D/E)
   ii. (A+B/D) / (E-F)+G
   iii. A*B- (C+D) / (E-F) +G/H
   iv. (X+Y-Z)/(H+K)*S
   v. J-K/G^H+(N+M)
   vi. A*(B-C)/D+E*F

3. Evaluate the following postfix expressions for the provided data.

   i. ab^c+d/e + where a = 5, b = 3, c = d = 2, e = 9
   ii. abcde+* +- where a = 12, b= 4, c = 7, d = 5, e = 2
   iii. ab + cd* + e* where a = 2, b = 6, c = 3, d = 5, e = 9

4. The efficient method used in evaluating a polynomial of the form

   $$P_n(x) = a_o r^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1}x + a_n$$

   is by nesting using Horner's rule, as shown below:

   $$P_n(x) = (\ldots (((a_o x + a_1)x + a_2)x + \ldots + a_{n-1}) \cdots)x + a_n$$

   Show how this can be carried out using a stack.

5. Consider the following arithmetic expression in postfix notation:

   752+*415-/-

   (a) Find the value of the expression.

   (b) Find the equivalent prefix form of the above expression.

   (c) Find the value of the expression from its prefix notation.

6. Convert the following infix expression into postfix expression.
   (i) (A-B) *X+ Y/ (F - C* E) + D
   (ii) X and Y OR NOT (A> B)

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7. Convert A + (B * C- (D/E -F * G) * H) into postfix form showing stack status after every step in tabular form.

8. There is a variation in the original Euclid's algorithm for computing the greatest common divisor of two integers M and N. According to the modified Euclid's algorithm,

$$GCD(M, N) = \begin{cases} GCD(M-N, N) & \text{if } M \geq N \\ M & \text{if } N = 0 \\ GCD(M, N-M) & \text{if } N > M \end{cases}$$

Using only a stack, write a procedure to compute the GCD as per the modified Euclid's method.

9. Consider the following stack of characters, where STACK is allocated N = 8 memory cells

STACK A, C, D, F, K,___,___,___

(For stack notational as the convenience, we use "___" to denote an empty memory cell.) Describe the following operations take place:

(a) POP(STACK, ITEM)
(b) POP(STACK, ITEM)
(c) PUSH(STACK, L)
(d) PUSH(STACK, P)
(e) POP(STACK, ITEM)
(f) PUSH(STACK, R)
(g) PUSH(STACK, S)
(h) POP(STACK, ITEM)

10. Consider the data in Problem 9. (a) When will overflow occur? (b) When will C be deleted before D?

11. Consider the following stack where STACK is allocated N = 4 memory cells:

STACK: AAA, BBB, ___,___.

Describe the stack as the following operations take place:

(a) POP(STACK, ITEM)
(b) POP(STACK, ITEM)
(c) PUSH(STACK, EEE)
(d) POP(STACK, ITEM)
(e) POP(STACK, ITEM)
(f) PUSH(STACK, GGG)

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12. Suppose the following stack of integers is in memory where STACK is allocated N = 6 memory cells:

    TOP =3 STACK: 5, 2,3,__,__,___

    Find the output of the following program segment:

    (i) Call POP(STACK, ITEMA).
    Call POP(STACK, ITEMB).
    Call PUSH(STACK, ITEMB + 2).
    Call PUSH(STACK, 8).
    Call PUSH(STACK, ITEMA + ITEMB).

    (ii) Repeat while TOP ≠ 0:
    Call POP(STACK, ITEM).
    Write: ITEM.
    [End of loop.]

13. Translate, by inspection and hand, each infix expression into its equivalent postfix expression

    (a) (A-B)/((D + E) * F)

    (b) ((A+B)/D) ↑ ((E - F) * G)

14. Let J and K be integers and suppose Q(J, K) is recursively defined by

$$Q(J,K) = \begin{cases} 5 & \text{if } J<k \\ Q(J-K,K+2)+J & \text{if } J>-=K \end{cases}$$

    Find Q(2,7) , Q(5,3) and Q(15,2).

15. Let A and B be nonnegative integers. Suppose a function GCD is recursively defined as follows:

$$GCD(A,B) = \begin{cases} GCD(B,A) & \text{if } A < B \\ A & \text{if } B = 0 \\ GCD(B, MOD(A,B)) & \text{otherwise} \end{cases}$$

    (Here MOD(A, B), read "A modulo B," denotes the remainder when A is divided by B.)

    (a) Find GCD(6, 15), GCD(20, 28) and GCD(540, 168). (b) What does this function do?

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16. Let N be an integer and suppose H(N) is recursively defined by

$$H(N) = \begin{cases} [3*N & \text{if } N<5 \\ 2*\ H(N-5) +7 & \text{otherwise} \end{cases}$$

a) Find the base criteria of H and (b) find H(2), H(8) and H(24).

17. Let M and N be integers and suppose F(M, N) is recursively defined by

$$F(M,N)= \begin{cases} 1 & \text{if } M = 0 \text{ or } M \geq N \geq 1 \\ F(M-1, N) + F(M-1, N-1) & \text{otherwise} \end{cases}$$

(a) Find (4, 2), F(1, 5) and F(2, 4). (b) When is F(M, N) undefined?

18. Let A be an integer array with N elements. Suppose X is an integer function defined by

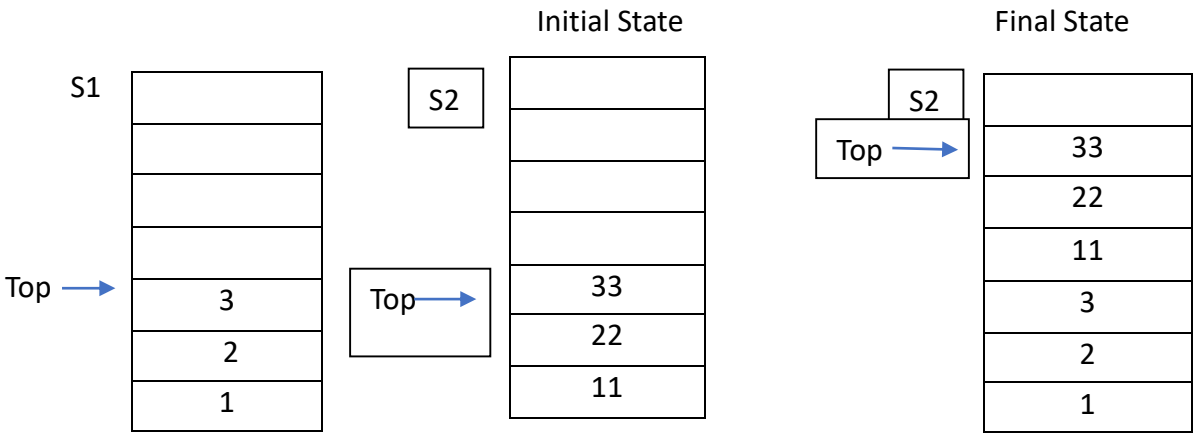
$$X(K) = X(A, N, K) = \begin{cases} 0 & \text{if } K = 0 \\ X(K-1)+ A(K) & \text{if } 0<K\leq N \\ X(K-1) & \text{if } K>N \end{cases}$$

Find X(5) for each of the following arrays:

(a) N= 8, A: 3, 7,-2, 5, 6, -4, 2, 7 (b) N = 3, A: 2, 7,-4

What does this function do?

19. Consider the stacks shown below:



Write a procedure that performs a series of push and pop operations on S_1 and S_2; so that the stack S_2 is transformed from its initial state to its final state.

20. Consider the following stack, where STACK is allocated N = 6 memory cells:

STACK: AAA, DDD, EEE, FFF, GGG,____

Describe the stack as the following operations take place:

(a) PUSH(STACK, KKK), (b) POP(STACK, ITEM),

(c) PUSH(STACK, LLL), (d) PUSH(STACK, SSS),

(e) POP(STACK, ITEM) (f) PUSH(STACK, TTT).

21. Suppose STACK is allocated N = 6 memory cells and initially STACK is empty, or, in other words, TOP = 0. Find the output of the following module:

1. Set AAA :=2 and BBB := 5.
2. Call PUSH(STACK, AAA).
   Call PUSH(STACK, 4).
   Call PUSH(STACK, BBB + 2).
   Call PUSH(STACK, 9).
   Call PUSH(STACK, AAA + BBB).

3. Repeat while TOP ≠0:
   Call POP(STACK, ITEM).
   Write: ITEM.
[End of loop.]
4. Return.

22. Consider the following stack of city names:
(a) STACK: London, Berlin, Rome, Paris,_____,_____
Describe the stack as the following operations take place:
   (i) PUSH(STACK, Athens),
   (ii) POP(STACK, ITEM)
   (iii) POP(STACK, ITEM)
   (iv) PUSH(STACK, Madrid)
   (v) PUSH(STACK, Moscow)
   (vi) POP(STACK, ITEM)

23. Write an algorithm to evaluate an arithmetic expression using stack and show how the expression 3*(5 - 3) will be evaluate.
24. Explain the two-stack problem in common area with all possible overflow and underflow conditions. Justify with an example.
25. Write a function or algorithm for converting decimal to binary using a stack.
26. Write a function or algorithm to reverse the string "HELLO" using a stack.

Submit on or before : 10 OCT. 2026